

[DC]Using Multimodal Input in Augmented Virtual Teleportation

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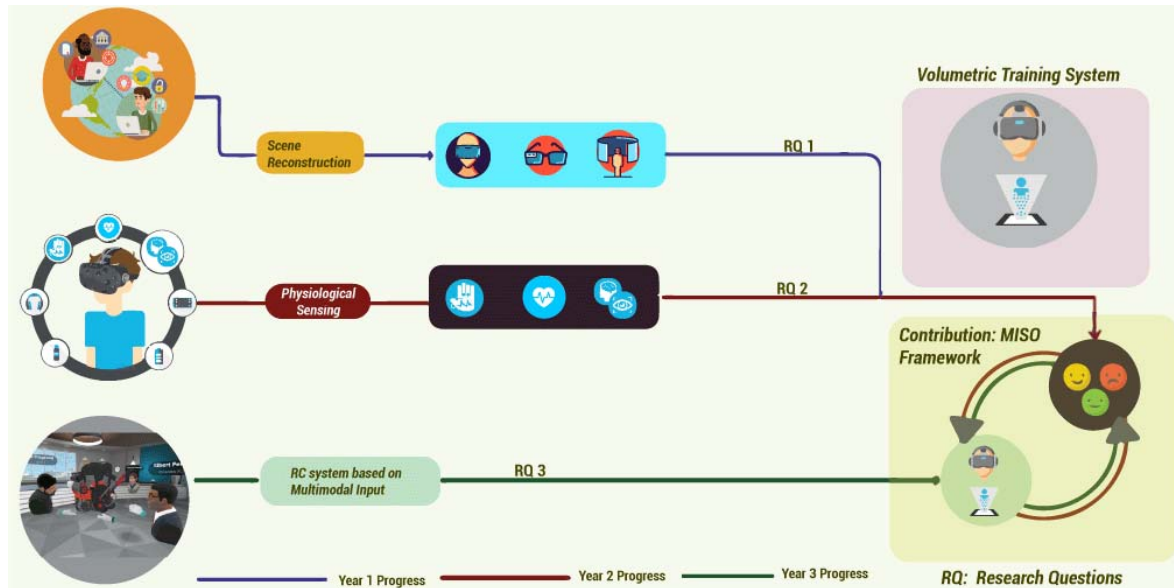


Figure 1: Thesis Overview

ABSTRACT

Augmented (AR) and Virtual Reality (VR) can create compelling emotional collaborative experiences, but very few studies have explored the importance of sharing a user's live environment and their physiological cues. In this PhD thesis, I am investigating how to use scene reconstruction and emotion recognition to enhance shared collaborative AR/VR experiences. I have developed a framework that can be broadly classified into two sections: 1) Live scene capturing for real-time environment reconstruction, 2) Sharing multimodal input such as gaze, gesture, and physiological cues. The main novelty of the research is that it is one of the first systems for real-time sharing of environment and emotion cues. It provides significant insight into how to create, measure, and share remote collaborative experiences. The research will be helpful in multiple application domains such as remote assistance, tourism, training and entertainment. It will also enable the creation of interfaces that automatically adapt to the user's emotional needs and environment and provide a better collaborative experience.

Index Terms: Human-centered computing—Human computer Interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Interaction design—Interaction design process and methods

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1 INTRODUCTION

Many communication cues are used in face-to-face collaboration, such as audio, visual, and environmental cues. To create more efficient remote communication, advances in telecommunication technology have enabled rapid developments in methods such as real-time audio and video streaming or mobile conferencing. However, these technologies cannot convey the same communication cues as in face-to-face, such as subtle gaze or gestures cues. It can also be challenging to share the spatial cues present in the local environment. The main contributions of my research is to address these limitations through the user-centered design, development, and evaluation of a multimodal-input based AR/VR collaboration system with real-time scene reconstruction and emotion sharing.

2 BACKGROUND

My work is based on earlier research in Volumetric capture, avatar representation, scene reconstruction, and sharing physiological cues. In this section I briefly summarise related previous research.

One of the key elements of our system is capturing and representing AR/VR remote collaborators. Researchers have tried several methods to capture and reconstruct the human body in 3D space [5]. For example, using volumetric playback to enhance the sense of presence in VR [2]. With the availability of affordable depth cameras, volumetric solutions like KinectFusion have emerged. Machine learning techniques have been used to predict skeletal joint positions from single RGB video [4]. Use of Voxels is another technique for enhancing visual coherence. With Holoportation [7], users wearing AR or VR displays could interact with virtual remote participants shown in the same physical space.

Avatar Representation is also important for my research. Representing the user's body improves social presence in remote collabor-

ration [10], and sense of spatial presence [9]. The use of Avatars for real-time communication has been explored in many applications. For example, [1] compared avatar representation preference across age groups and found that older adults prefer attractive avatars with expressive features. Considerable research has been done on how the appearance of avatars influences communication and interaction [6].

While earlier research has focused on creating and measuring emotions in VR, little work has been done to investigate the effects of providing physiological feedback to collaborators, and measuring its effects on usability, immersion and interaction. Dey et al. [3], reported that showing real time heart rate to collaborators could create higher subjective connections in a collaborative VR setup.

Compared to this earlier work, the main novelty of my research is developing and evaluating a multimodal-input remote collaboration system that shares the live environment of the collaborators. The system also shares the heart rate, cognitive load, attention and relaxation states of the collaborator.

3 CURRENT AND FUTURE WORK

We have developed a remote collaboration system to capture and share the live surroundings of a local user with a remote expert, and we conducted a user study to evaluate the effect of sharing hand gesture and eye gaze cues.

3.1 Live Reconstruction

We developed an inside-out live 3D panorama sensor cluster which supports instant 3D reconstruction with real-time updating. This enables the remote expert to see the entire local user's workspace in 3D and receive all updates without delay. The unit has large scale coverage (10m radius) in a small footprint. The remote expert's gaze and gesture is detected by the VR headset's gesture sensor and built-in eye tracker, and is shared seamlessly to the local worker's. This is for showing task guiding and attention focus, and offering rich natural communication cues.

The application we are focusing on is training. Conventional teleconferencing training systems typically allow users to see each other's faces, while sharing 2D content like slides. However, these methods lack depth information and a free 3D perspective of the training content. We investigated the impact of volumetric playback in a Mixed Reality (MR) spatial training system. Building upon our previous research, four spatial instruction cues were explored; *Annotation*, *Hand gestures*, *Avatar*, and *Volumetric playback*. Through two user studies simulating a mechanical assembly task, we found that the volumetric visual cues enhanced spatial perception, exhibiting increased co-presence and system usability while reducing mental workload and frustration. We also found that the tasks required less effort and mental load when eye gaze was incorporated, but gaze on its own was not perceived to be very useful [8].

4 EXPECTED CONTRIBUTIONS

The research contributions of my proposed dissertation to the field of human-computer interaction, affective computing, and remote collaboration could be categorized in two primary classes:

4.1 Artifact Contributions

[1] An Empathic remote collaboration framework; a Unity plugin that integrates various physiological sensors and design guidelines for remote collaboration. The final research outcome will be a context-aware, emotion adaptive VR environment that will use the user's real-time physiological information for remote collaboration. [2] A training system that supports various visualizations of the collaborator/trainer like volumetric reconstruction, avatar representation, hand gestures, annotations etc, to enhance the sense of co-presence. [3] Live scene reconstruction system based on a local worker (AR) and remote user (VR).

4.2 Empirical Contributions

[1] Experimental design methods in VR/AR remote collaboration to evaluate immersion, cognitive load and co-presence based on various visualizations of the collaborator. [2] Methods to share the collaborator's physiological cues to provide an enhanced sense of co-presence and immersion. [3] Findings from user studies conducted using a multimodal sharing framework to contribute further empirical knowledge to develop design guidelines and answer research questions for sharing multimodal input in remote collaboration with live scene reconstruction.

5 EXPECTED BENEFITS OF PARTICIPATION

Participation in the doctoral consortium will benefit me in the following ways. Having discussions with experts and consortium participants that could suggest research methods to use, exchange of ideas for future work, and recommendations on other past research work. Advice from expert researchers from the IEEEVR community will be invaluable and will strengthen my dissertation. I will also contribute to discussions with fellow consortium participants. I am happy to share my expertise with live scene reconstruction, volumetric training, physiological sensing, analysis of physiological data, and AR/VR remote collaboration. I also look forward to forming collaborations which could enhance my further research in the field.

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